
Faster, Profitable, Yet Peripheral: The Role of Shared E-Scooters in Urban Mobility

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Abstract

Privacy regulations prevent e-scooter operators from releasing trip data, limiting systematic analysis of this growing urban mobility mode. We reconstruct 2.15 million trips across 26 cities by matching vehicle disappearance and reappearance events from public availability feeds, using distance, speed, and battery priors. Four findings emerge: e-scooter sharing achieves 44% profit margins but is highly sensitive to trip matching confidence; cities with slower public transit do not adopt e-scooters more heavily, suggesting factors beyond time savings drive usage; weather effects are significant, with wind chill and precipitation jointly explaining 14% of ridership variance; and weekday demand shows a universal dual-peak commuting signature regardless of city size. Shared e-scooters appear operationally viable but complementary with existing transit infrastructure.

1. Introduction

Since their legalization in Germany in June 2019 (Bundesgesetzblatt, 2019), e-scooters have been marketed as sustainable alternatives for short-distance travel. However, e-scooters have attracted negative attention over safety concerns and sidewalk clutter (Wallgren et al., 2023), and surveys consistently find that they predominantly replace walking and public transit rather than car journeys (Weschke et al., 2022; Laa & Leth, 2020). These negative claims question the necessity of e-scooters in cities, yet systematic analyses of real-world usage data remain scarce.

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The open-data initiative MobiData BW (MobiData BW, 2022) offers an opportunity to close this gap by aggregating real-time vehicle availability feeds from multiple operators across Baden-Württemberg. However, the GBFS standard enforces resetting vehicle identifiers after each trip to protect privacy, so individual rides must be *inferred* from disappearance and reappearance events.

This paper investigates whether rental e-scooters constitute a viable component of urban mobility infrastructure and factors that drive e-scooter usage. Real-time availability data was collected from the MobiData BW API and a **matching algorithm** was used to infer individual rides from observed events. Based on this reconstructed trip data, we **assessed economic viability** by modeling the revenue structure and operational costs of e-scooter sharing services. We then **examined intermodal relationships** between e-scooter usage and public transportation (PT) to determine whether these services complement or compete with existing transit infrastructure. Finally, we **identified external determinants** of ridership, including weather, temporal and spatial characteristics.

2. Data Collection and Processing

We set up a polling device to query the MobiData BW API at one-minute intervals and store vehicle events whenever significant changes are detected. These include location changes, battery level changes, or status changes, such as appearing or disappearing.

Not all disappearances/appearances correspond to ride starts/ends; vehicles also vanish due to maintenance or operator relocations. We attempt to filter such events by flagging patterns indicative of fleet operations (e.g. reappearance with fully charged batteries) and excluding them from the matching process. For the remaining candidate pairs, we compute a log-likelihood score:

$$\mathcal{L} = \log P(d) + \log P(v \mid d, \Delta t) + \log P(r \mid d) \quad (1)$$

where d is the routed distance between start and end locations (obtained via the Valhalla routing engine using OpenStreetMap (OSM) data (Valhalla Contributors, 2024; Open-

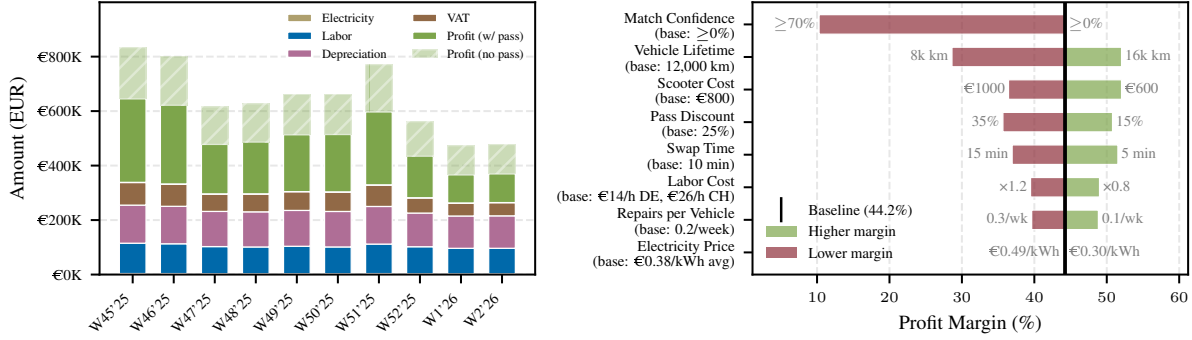


Figure 1. Profitability analysis of e-scooter operations. (Left) Weekly revenue decomposition into operational costs and profit. Solid green represents conservative profit estimates assuming a 25% revenue reduction from subscription passes; hatched regions indicate additional profit under full pricing. (Right) Tornado diagram showing profit margin sensitivity to key assumptions. Bars indicate the margin range when varying each parameter individually while holding others at baseline (vertical line, 44.2%).

StreetMap Contributors, 2024)), $v = d/\Delta t$ is the implied travel speed, and r is the observed range consumption. The distance prior $P(d)$ follows a log-normal distribution reflecting typical urban trip lengths of 1–2 km (Li et al., 2022; Krauss et al., 2024; Reck et al., 2021); the speed likelihood $P(v | d, \Delta t)$ is a truncated normal between 5–23 km/h centered at 14 km/h, consistent with average speeds of 10–15 km/h in European cities (Li et al., 2022; Degele et al., 2022); and the range likelihood $P(r | d)$ assumes consumption scales linearly with distance at approximately $1.1 \times$ efficiency. Expectation-Maximization refinement of the parameters proved unreliable: convergence depended heavily on initialization, and estimates drifted toward unvalidatable local minima. We therefore retain literature-grounded priors, with final assignments computed via softmax normalization.

3. Operational profitability

Operational profitability represents a minimum viability threshold: if e-scooter sharing cannot cover direct operating costs, the business model is fundamentally unsustainable. We assess this using a cost model based on observational data, in-app pricing, and industry benchmarks.

Selecting the best match per disappearance (median confidence 77.7%, IQR 52.3–96.4%) and applying provider pricing (0.50–1.00 € unlock plus 0.20–0.50 €/min), we estimate 6.99million€ net revenue from 2.15million trips, reduced to 5.25million € with a 25% pass discount. Operating costs total 2.93million€: depreciation (31%), labor (24%), electricity (<1%). This yields a 44.2% profit margin, though profitability varies by provider (Figure 1a).

Sensitivity analysis (Figure 1, right) reveals that match confidence has the largest impact: requiring $\geq 70\%$ confidence removes 43% of trips, dropping the margin to 10.3%. Vehicle lifetime (8–16kkm) and acquisition cost (600–1000€)

each swing margin by 15–23pp; pass discounts (15–35%) contribute 15pp. Labor and maintenance costs show lower sensitivity as they scale with volume. These results suggest that e-scooter businesses do clear the hurdle of operational viability, though indirect costs such as software development, marketing, and administration remain unobserved.

4. Interaction with Public Transportation

Whether e-scooters complement or compete with public transit remains debated. Surveys suggest 30–50% of trips replace walking or PT (Weschke et al., 2022; Laa & Leth, 2020), yet how public transit quality influences e-scooter adoption is underexplored.

To investigate this relationship, we first computed PT travel times for each matched e-scooter trip using the MOTIS routing engine (MOTIS Project, 2024). For 1.87 million trips with valid PT routes, public transport required on average 35.0 minutes compared to 15.0 minutes by e-scooters. This average advantage of 20 minutes was more pronounced during night hours (21:00–06:00), when reduced service frequencies increased average PT journey times.

We hypothesized that cities with slower PT would exhibit higher e-scooter adoption rates. To test this, we calculated daily rides per 1,000 inhabitants for 23 cities and correlated this metric with the average PT time disadvantage. Figure 2 displays the results separately for daytime and nighttime trips, with point sizes scaled by trip volume.

Contrary to our hypothesis, we found no significant positive correlation between PT time disadvantage and e-scooter usage. Spearman correlations were weakly negative for both day ($\rho = -0.18$, $p = 0.41$) and night ($\rho = -0.28$, $p = 0.21$) periods. This suggests that factors other than relative transit speed such as tourism, urban density, first-/last-mile connectivity, or provider coverage

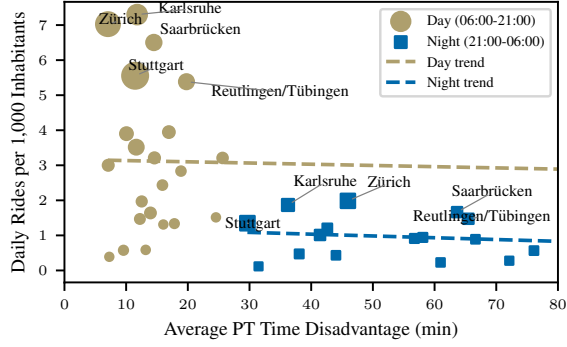


Figure 2. E-scooter usage versus PT time disadvantage across 23 cities. Point size indicates trip volume.

increase e-scooter adoption. Nonetheless, e-scooters represent a negligible fraction of urban mobility. Compared to official PT ridership (Stuttgarter Straßenbahnen AG, 2024; Zürcher Verkehrsverbund, 2024; Verkehrsverbund Neckar-Alb-Donau, 2024), e-scooters account for <1% of transit volumes: 0.68% in Stuttgart, 0.25% in Zürich, and 2.26% in Reutlingen/Tübingen. Shared e-scooters operate at too small a scale to substitute for PT. This raises the question: if faster travel times do not boost adoption in cities with slower PT, what factors do drive or hinder e-scooter usage?

5. Weather Effects on Scooter Usage Behavior

One clear advantage of PT over e-scooters is weather independence. We next quantify the extent to which ridership declines under adverse conditions. Prior work confirms weather sensitivity in e-scooter demand (J. K. Mathew & D. M. Bullock, 2024), and the revenue drop during colder periods is visible in our data (Fig. 1, left). To collect weather data we used Open-Meteo API (Open-Meteo, 2026). Data

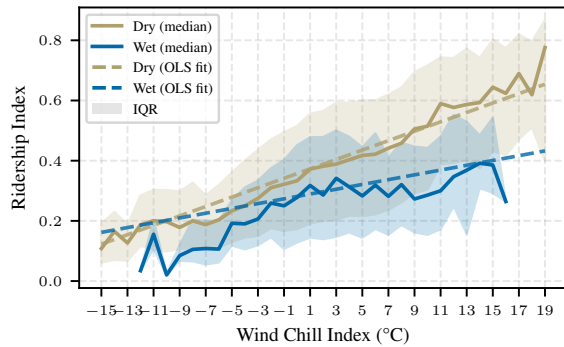


Figure 3. Normalized E-scooter Usage versus WC Index and Wet and Dry Weather Conditions. $WindChill_F = 35.74 + 0.6215T_F - 35.75V^{0.16} + 0.4275T_FV^{0.16}$, where: T_F - air temperature in °F, V - wind speed in mph, adapted to metric system.

were parsed hourly for each major city in the dataset throughout the study period. To assign weather conditions to each ride, we used a K-d Tree (Bentley, 1975) algorithm to look for the geographically closest weather station. Based on our preliminary analysis, we focused on temperature, wind speed, and precipitation (rain and snow). To capture how cold weather ‘feels’, we implemented a Wind-Chill (WC) index (USA National Weather Service, 2025), which combines both temperatures and wind speed.

Cities with demand surges from new provider entries were excluded. Trip counts were aggregated hourly and transformed into a Normalized Ridership Index by dividing by the maximum count for that hour across the study period. Precipitation was grouped into wet (>0.1 mm) and dry conditions (American Meteorological Society, 2024).

$$I = \beta_0 + \beta_1 \cdot WC + \beta_2 \cdot Wet + \beta_3 \cdot (WC \times Wet) \quad (2)$$

To quantify the effects of weather conditions, an Ordinary Least Squares (OLS) regression was performed. The model explains 13.7% of the variance ($R^2 = 0.137$). All coefficients are statistically significant ($p < 0.001$, $N = 41,324$). **WC** ($\beta_1 = 0.0156$): Under dry conditions, a 1-degree increase in WC raises the ridership index by 1.56 percentage points. **Wet** ($\beta_2 = -0.0756$): At freezing temperatures (0°C), wet weather reduces demand by 7.6 percentage points compared to dry conditions. **Interaction** ($\beta_3 = -0.0076$): Precipitation weakens the positive effect of temperature. While dry days show a 1.6 percentage point gain per degree, wet days show only a 0.8 percentage point gain.

Fig. 3 shows the positive correlation between ridership and the WC index (0.35 for dry; 0.20 for wet). Wet weather suppresses demand regardless of temperature, with a wider gap observed in warmer temperatures (>5°C) and a narrower gap in colder conditions (<3°C). This disparity might suggest a selection effect. In extreme cold sensitive users have likely opted out of riding an e-scooter. While the correlation between ridership and temperature is positive, weather does not fully explain ridership fluctuations, suggesting other influential factors. Furthermore, since the coldest snowy weather coincided with the winter holiday season, the observed weather effects may be contaminated with the naturally lower demand typical of the holidays.

6. Usage Pattern Characterization

Sections 4 and 5 established that neither public transit quality nor weather fully explains e-scooter adoption. To better characterize the functional role e-scooters play in urban mobility, we analyze temporal demand patterns and spatial trip flows. Specifically, we ask whether usage reflects utilitarian commuting or discretionary leisure.

Figure 4 (left) displays z-score normalized hourly trip counts across 19 cities with more than 5,000 trips to filter out high

variance, low ridership cities. Weekday usage exhibits a characteristic dual-peak structure aligned with commute hours: a morning peak (7–9h) and a more pronounced evening peak (17–19h). Weekend patterns differ markedly, showing a single broad afternoon peak (14–18h) consistent with leisure activities. The temporal signature is consistent across cities of varying size.

We retrieved buildings within 100m of each origin and destination from OSM ([OpenStreetMap Contributors, 2024](#)) and classified them into functional categories. We computed a flow metric as the difference between destination and origin building type proportions (Figure 4, right). The results align with the temporal patterns: weekday mornings show outflow from residential areas and inflow to commercial and educational areas; evenings reverse this pattern. Weekends show increased flow toward entertainment and recreation venues. PT stations consistently appear as destinations in the morning and as departures in the evening, suggesting a first- and last-mile role.

However, these spatial patterns should be interpreted cautiously: designated parking zones constrain drop-off locations, and dense urban environments mix Building types within short distances. Nevertheless, the alignment between temporal peaks and spatial flows indicates that weekday e-scooter usage serves utilitarian commuting purposes, while weekend usage skews toward leisure.

7. Discussion & Conclusion

Our findings paint a nuanced picture of e-scooters’ role in urban mobility. The demonstrated operational viability guarantees survivability, but not broader adoption: despite offering substantial time savings over public transit, e-scooters remain a marginal mode of transportation with less than 1% of daily drivers compared to PT. The absence of correlation

between PT speed disadvantage and adoption rates suggests that e-scooter use is driven less by rational time optimization than by convenience, habit, or contextual factors we cannot observe in availability data.

Several limitations constrain these findings. The analysis covers autumn and winter in predominantly German cities; summer demand and battery behavior remain unexplored. Trip reconstruction introduces uncertainty that propagates through ride counts, trip lengths, and ultimately revenue estimates. Spatial distributions around departures and arrivals suggest broad usage patterns but cannot quantify actual intent. Overcoming these challenges would require provider-level access to ride statistics and app usage data.

Key questions remain beyond our data’s reach. Per-minute pricing may deter rides where time savings are greatest. App-based access introduces friction that PT cards do not. Comfort and safety concerns may discourage risk-averse users. E-scooters may compete more directly with bicycles than with buses or trams. Disentangling these factors requires survey data that trip reconstruction cannot provide.

The matching algorithm itself offers room for refinement. Sparse nighttime traffic yields near-deterministic but biased matches that could inform improved priors. Battery consumption likely varies with temperature and elevation; speed distributions shift by time of day. Modeling rides concurrently and allowing group riding by multiple riders on the same route could improve reconstruction accuracy beyond the current 77.7% median confidence and broaden matching beyond restricted priors, making it more flexible.

Overall, e-scooters provide a viable mode of transportation and exhibit regular commuting and leisure patterns, indicating a dedicated core user group. Yet their role in urban mobility remains peripheral as a convenient option for some, but far from a structural shift in how cities move.

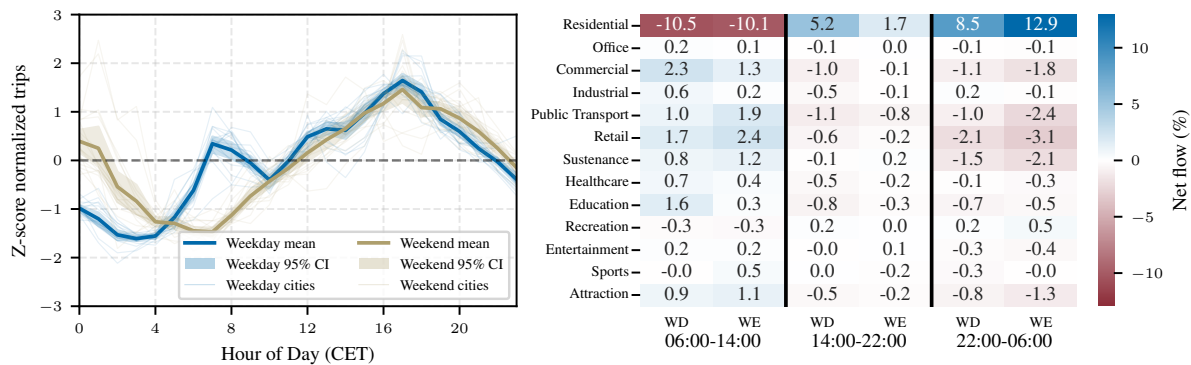


Figure 4. Temporal usage patterns and drive intentions of shared e-scooters. **(Left)** Z-score normalized hourly trip counts comparing weekday (blue) and weekend (gold) patterns. **(Right)** Drive intention flow computed as the difference between destination and origin building type proportions across three time phases. Positive values (blue) indicate net inflow, negative values (red) indicate net outflow.

Contribution Statement

A. Sarymsakova researched the cost model and implemented the business model. E. Ehbrecht designed the research structure and led the data collection and data processing. F. Tubaro conducted the analysis of temporal and spatial usage patterns. M. N. Yildirim conducted the public transportation analysis. P. Kripitsa was responsible for the collection and analysis of the weather data. All code and data are publicly available at:

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